Pairs trading on KO/PEP: Hackathon report

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1 AI usage disclosure

AI assistance was used on the coding side to generate code snippets, support debugging, clarify the usage of specific Python libraries and accelerate implementation tasks. On the writing and research side, it was used to assist the discovery of potentially relevant tests and datasets. In all cases, however, AI remained a supportive and non-autonomous tool. The mathematical interpretations, experimental design and final presentation of results were independently checked and validated by me. No conclusion was included in this report solely on the basis of AI-generated output. What I mainly checked or corrected myself:

- I asked to remove the out-of-sample data from the Phase 1 plots, which the first version included.
- I rejected extra structure I did not need and chose not to add statistical checks I could not explain myself.

Models used: *Claude Opus 5.5*

2 Writeup

I chose the pair KO/PEP because I know the sector better than XOM/CVX and also because they operate in a more mature market than INTC/AMD.

I downloaded the dividend-adjusted daily prices from 1990 to October 2026 and, before looking at any chart, split them into in-sample (1990-2014) and out-of-sample (2015-2026, about 30%). For the exploration of the data in phase 1 I used only the in-sample data. During the second phase I estimated the hedge ratio with OLS (KO as dependent variable) and checked whether the spread is mean-reverting by comparing its mean crossings with 1,000 random walks and it showed no evidence of mean reversion. The baseline (60-day z-score, entry at 2σ, exit at 0, 5 bp costs) earned a Sharpe of 0.45 in-sample, mostly in 1999-2004. I tried 12 in-sample configurations in total (9 planned, 3 added after seeing the grid), all logged with timestamps, and did not change the baseline. On the out-of-sample, run once, the Sharpe was -0.50.

For phase 3 I chose the medium method (rolling hedge ratio and rolling Engle-Granger test), because it checks directly whether the relationship behind the spread stays stable over time. The test doesn't find evidence of cointegration on the whole in-sample ($p = 0.363$), and only short, scattered windows where the test rejects. The phase 2 strategy on the periods flagged a strong even loses money.

For phase 4 the Nasdaq source only covers PEP, so I used FINRA dataset for both stocks, short interest showed no link with the pair.

Finally for phase 5 I chose to use a joint block bootstrap with 10-day blocks, which removes any mean reversion beyond the block length, rather than an Ornstein-Uhlenbeck process, which would mean-revert by construction. The strategy also made money on simulated histories without mean reversion, and the real Sharpe was beaten by 6% of them.

After Phase 5 I did not add new methods. I used the remaining time to review the conclusions, add the explanations and the maths behind the tests to the notebook, and rerun everything from scratch to check every number in this report.

Conclusion. The in-sample profit comes mostly from fitting the hedge ratio on the same data, from KO and PEP simply moving together over a few days, and from one lucky period, not from a real, tradable relationship between the two stocks.

With more time, I would repeat the analysis on other pairs to see whether these results generalise.